

Algorithmic accuracy is not clinical readiness in fetal cardiac screening



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Artificial intelligence (AI) has entered fetal cardiac ultrasound with performance metrics that increasingly approach expert-level discrimination in curated retrospective datasets.^{1,2} High area under the curve values for plane recognition and selected anomaly detection have fueled enthusiasm that AI may improve prenatal screening for congenital heart disease (CHD).^{1,2} Yet technical performance should not be mistaken for clinical readiness. For population-based fetal cardiac screening, the threshold for adoption is higher: a model must perform reliably in low-prevalence, heterogeneous, real-world environments and demonstrate clinical utility beyond retrospective accuracy. This perspective is not intended to suggest that AI is poised to replace clinicians in fetal cardiac imaging, but rather to emphasize that growing enthusiasm for promising algorithmic performance should not be mistaken for evidence of readiness for clinical implementation. In this sense, our aim is less to anticipate imminent replacement than to provide a timely clinical pause for reflection before technical enthusiasm outpaces evidentiary standards.

This distinction is especially important in routine second-trimester obstetric screening.³ In unselected populations, CHD prevalence is low, and even strong sensitivity and specificity may translate into modest positive predictive value.^{1,3} In practice, this means that a substantial proportion of positive alerts may represent false positives, with consequences that matter in maternal-fetal medicine: increased referrals, parental anxiety, added downstream imaging, and inefficient use of specialized resources. At the same time, small reductions in sensitivity may result in missed serious anomalies. Screening settings are therefore unforgiving to even modest

performance degradation. Obstetrics has previously experienced the consequences of premature technological adoption, most notably with electronic fetal monitoring, which was widely implemented before robust evidence demonstrated meaningful improvement in neonatal outcomes. AI in prenatal imaging should avoid a similar trajectory, in which intuitive appeal and early enthusiasm precede rigorous demonstration of clinical benefit.

Fetal cardiac imaging is also intrinsically variable.³ Image quality and interpretability depend on gestational age, fetal lie, maternal body habitus, ultrasound equipment, acquisition settings, and operator skill and experience. These factors differ substantially across centers and between community and tertiary environments. A model trained in relatively homogeneous datasets may not perform similarly when deployed across different institutions, vendors, and sonographer experience levels. This problem of domain shift is well recognized across medical imaging⁴ and is particularly relevant in obstetric ultrasound, where acquisition variability is fundamental to the examination itself. Generalizability should therefore be demonstrated, not assumed. This issue becomes especially important in fetal imaging, where the variability arises not only from patient and device characteristics, but from the acquisition process itself, making external validation across heterogeneous settings a prerequisite for clinical credibility.

Dataset construction is equally important. In fetal cardiac ultrasound, labels are not simple ground truths; they depend on anatomical expertise, image interpretation, diagnostic definitions, and often consensus adjudication. Variability in plane definition, annotation thresholds, and outcome confirmation can introduce systematic bias into training and testing datasets. If these processes are not transparent, models may inherit hidden bias while appearing technically robust. This concern is compounded by the black box nature of many AI systems, in which the reasoning underlying model outputs remains opaque to the user. Such opacity limits interpretability, complicates reproducibility, and may undermine clinical trust, particularly when models appear highly performant but provide little insight into how conclusions are reached. Reporting standards such as TRIPOD-AI and tools such as PROBAST-AI provide an important framework for evaluating reproducibility, applicability, and risk of bias^{5,6}. In this context, dataset quality is not a peripheral technical issue but a prerequisite for trustworthy clinical translation.

Another limitation is that most available fetal cardiac AI studies remain retrospective.^{1,2} Demonstration of discrimination in archived datasets is an important early step, but it does not establish benefit in practice. For screening tools,

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clinically meaningful evaluation should include prospective evidence that AI improves detection, referral appropriateness, workflow efficiency, or patient-centered outcomes in unselected populations. Without such evidence, claims of readiness for broad implementation remain premature.

Human-AI interaction also deserves more attention. AI does not enter a neutral environment; it changes how clinicians scan, interpret, and decide. Automated prompts may improve vigilance in some users, but they may also increase alert fatigue, shift referral thresholds, or encourage overreliance on the system. These workflow effects are especially relevant in prenatal care, where downstream consequences extend beyond diagnostic accuracy alone. A system that performs well technically may still fail clinically if it increases unnecessary escalation or erodes operator judgment. It also involves contextual judgment, communication of uncertainty, parental counseling, and empathic interaction—dimensions of care that remain fundamentally human and are not replicable by algorithmic systems. This human dimension is not peripheral to fetal imaging, but central to its clinical practice, and represents a fundamental boundary to full automation. In this context, AI should be understood not as a substitute for clinical expertise, but as a support tool whose value depends on how safely and thoughtfully it is integrated into human care. Equally important, prenatal diagnosis is not limited to anomaly recognition.

AI in fetal cardiac ultrasonographic assessment is promising and may ultimately help standardize image acquisition, reduce operator dependency, and improve access to high-quality screening. Although some emerging tools aim to assist with image acquisition through real-time guidance or quality assessment, image acquisition remains fundamentally operator dependent, and current claims of clinical readiness relate primarily to AI-assisted interpretation and decision support rather than fully autonomous scanning. However, enthusiasm for innovation should not compress the evidentiary pathway from algorithmic performance to clinical adoption. Before integration into population-based fetal cardiac screening, AI systems should demonstrate robust external validation across heterogeneous settings, transparent dataset and annotation methods, evaluation in true low-prevalence screening cohorts, and prospective evidence of clinical benefit.^{3,5,6} Clinical readiness should also include regulatory oversight, ethical

safeguards, legal accountability, and equitable implementation across diverse patient populations and care settings. These are not secondary considerations, but core requirements for responsible clinical adoption. In prenatal care, where disparities in access to expertise are already substantial, AI may either reduce inequities or reinforce them depending on how it is designed, validated, and deployed.

Algorithmic accuracy is a technical achievement. Clinical readiness in fetal cardiac screening is a separate, and higher, standard.

Declaration of AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT to edit the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication. ■

CRediT authorship contribution statement

Edwin Quarello: Writing – original draft, Conceptualization. **Sifa Turan:** Writing – review & editing.

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